

CSCI 240 -- PA 3

Vectors, Lists, & Iterators

Feel free to discuss and help each other out but does not imply that you can give away your code or your answers! Make sure to read all instructions before attempting this lab.

You must use an appropriate provided template from Canvas and output "Author: Your Name(s)" for all your programs. If you are modifying an existing program, use "Modified by: Your Name(s)".

Exercise 1: Use a vector (C++ **ArrayVector** or Java **ArrayList**) from the textbook for this exercise. **Modify the class to add operation removeElem(str)** and then use it to perform operations on a vector of strings. The removeElem() operation removes an item if it is in the list and does nothing if it is not in the list. Set up a driver to create a vector, add a few items, remove a few items, and print the vector. Just include the driver and removeElem() operation in the submission of your source code.

Try the following:

1. Add "Two" to index 0
2. Add "Three" to the 1
3. Add "One" to index 0
4. Add "Four" to index 3
5. Print the vector
6. Remove item at index 0
7. Remove "Two"
8. Remove "four"
9. Insert your name at index 1
10. Print the vector

Exercise 2: Use a list (C++ **DoublyLinkedList** or Java **LinkedPositionalList**) from the textbook to perform operations on a list of strings. Set up a driver to create a list, add a few items, remove a few items, and print the list. You must declare and use an iterator in your driver. Just include the driver in the submission of your source code.

Try the following:

1. Add "Two" to the front
2. Add "Three" to the rear
3. Add "One" to the front
4. Add "Four" to the rear
5. Use an iterator to print the list
6. Set up an iterator at the begin and advance twice
7. Insert your name before the iterator

8. Remove front
9. Remove back
10. Use an iterator to print the list

Exercise 3 (more points for this exercise): Set up **TextEditor** class, which can be used to store, edit, and display a string of characters using a list implementation from exercise 2. The editor (TextEditor class) uses either C++ **DoublyLinkedList** or Java **LinkedPositionalList** where each node stores only 1 character and it must support the following operations:

- left() – move cursor left one character (do nothing if at beginning).
- right() – move cursor right one character (do nothing if at the end).
- insert(c) – insert the character c just before the cursor.
- remove() – remove the character just before the cursor (like backspace if there is at least one character before it).
- first() – move cursor to the beginning (before the first character)
- last(i) – move cursor to the end (after the last character).
- display() – display string (including spaces and cursor).

Use character | for the cursor so if your characters are "Hello World" and the cursor at the beginning then "|Hello World" (i.e, the cursor is before character 'H' in the example). The cursor shall not be stored in the list but should be printed on the screen to indicate where the cursor is at that moment. *Note: most operations will be based on the current position of the cursor which is just before the iterator and the first 6 operations would run in $O(1)$.* We will only allow letters in the alphabet and space as valid characters. When an editor object is created, it can be an empty string as a default option, or you can specify a string such as "Hello World" (assume that the cursor will be at the end of the string).

Set up a driver to create a TextEditor object, then use it to perform a few operations and then print the text. Try the following cases below. Just include the driver and class TextEditor in the submission of your source code.

1. Create a TextEditor object with "Hello Word" (assume cursor at the end)
2. Display current information (should be "Hello Word|")
3. Move cursor right (nothing should happen)
4. Move cursor left
5. Insert '1' (digit 1, ignore invalid character)
6. Insert 'l' (letter l)
7. Display current information (should be "Hello Worl|d")
8. Move cursor to the beginning
9. Move cursor right
10. Remove character
11. Move cursor to the end
12. Display current information (should be "Hello World|")

Question 1: Would you use an array to implement List ADT? Explain why or why not.

Question 2: Given a choice between a vector and a list, when should you select a vector over a list?

Extra Credit: Set up **TextEditor1** class, which can be used to store, edit, and display a string of characters using a string. The editor (TextEditor1 class) uses either C++ **string** class or Java **String/StringBuilder** class and it must support the same operations in exercise 3. Run the same test cases as well.

Fill out and turn in the PA submission file for this assignment (save as PDF format).